Overview

The **SPARK SR1120** is a digitally programmable UWB (Ultra-Wideband) wireless transceiver RFIC implementing SPARK LE-UWB™ technology. With an operating frequency range of 6.2 GHz to 9.5 GHz, the SR1120 provides ultra-low-power, deterministic wireless communications with short, predictable latency at symbol rates up to 40.96 Msymbol/s. The SR1120 is optimized for energy-efficient operation across a wide range of data rates while maintaining a robust wireless link, including in RF-congested environments.



Figure 1: SR1120

The SR1120 integrates an RF transmitter and receiver, antenna-diversity support, digital baseband hardware, a power-management unit, a low-power clock input, and a sleep counter. A host MCU configures and controls the device through a standard SPI or QSPI interface, simplifying integration into compact, ultra-low power wireless systems.

The impulse-radio architecture combines short-duration transmissions with aggressive duty cycling for minimal energy consumption across a wide range of operating conditions. Digitally programmable output-spectrum shaping enables operation within multiple spectral masks and supports compliance with worldwide UWB regulatory emission requirements. The SR1120 supports both energy-efficient data communications and ranging applications, making it ideal for a wide range of consumer, industrial, medical, always-on IoT, and next-generation advanced datacom applications.

Features

- UWB Radio Transceiver
  - 6.2 – 9.5 GHz operating frequency range
  - Configurable symbol rate: 20.48, 27.3067 and 40.96 MHz PHY rates
  - Tunable TX symbol energy of up to 0 dBm / 20.48 MHz
  - RX sensitivity (symbol energy) of -81 dBm / 20.48 MHz
  - Antenna diversity support
- Integrated clock management
  - Only one 32.768 KHz external XTAL required
  - Timer always available except in shutdown mode.
  - Synthesized 40.96 MHz system clock.
- Versatile digital core
  - Up to 40 MHz SPI and Quad-SPI interfaces
  - Independent 256 Bytes transmit and receive buffers
  - Modem supports OOK modulation, differential encoding, 32-bit sync word, punctured FEC, 16-bit CRC
  - 1 Kbit NVM
- Ultra-low power consumption
  - 1.8 to 3.3 V supply
  - Internal DC-DC converter
  - Energy efficiency of 0.3 nJ/bit for TX and 0.7 nJ/bit for RX
- Ultra-low latency
  - 25 us airtime for 1 Kbit at a PHY rate of 40.96 MHz
  - Ideal for real-time data streams such as Headset and HID applications
- Coexistence and reduced interference with BLE, WIFI (2.4, 5 & 6 GHz) and cellular
- Meets JEDEC JESD471, JESD22 industry standards
- Industrial operating range: -40 to +85 ℃
- Package
  - 4 mm x 4 mm 32 pin QFN
  - WLCSP ready
- Low-cost BOM
- Support for Carrier-Sense (CSMA) stack
- SDK support
- Evaluation kits available

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1 System Overview

1.1 Block diagram

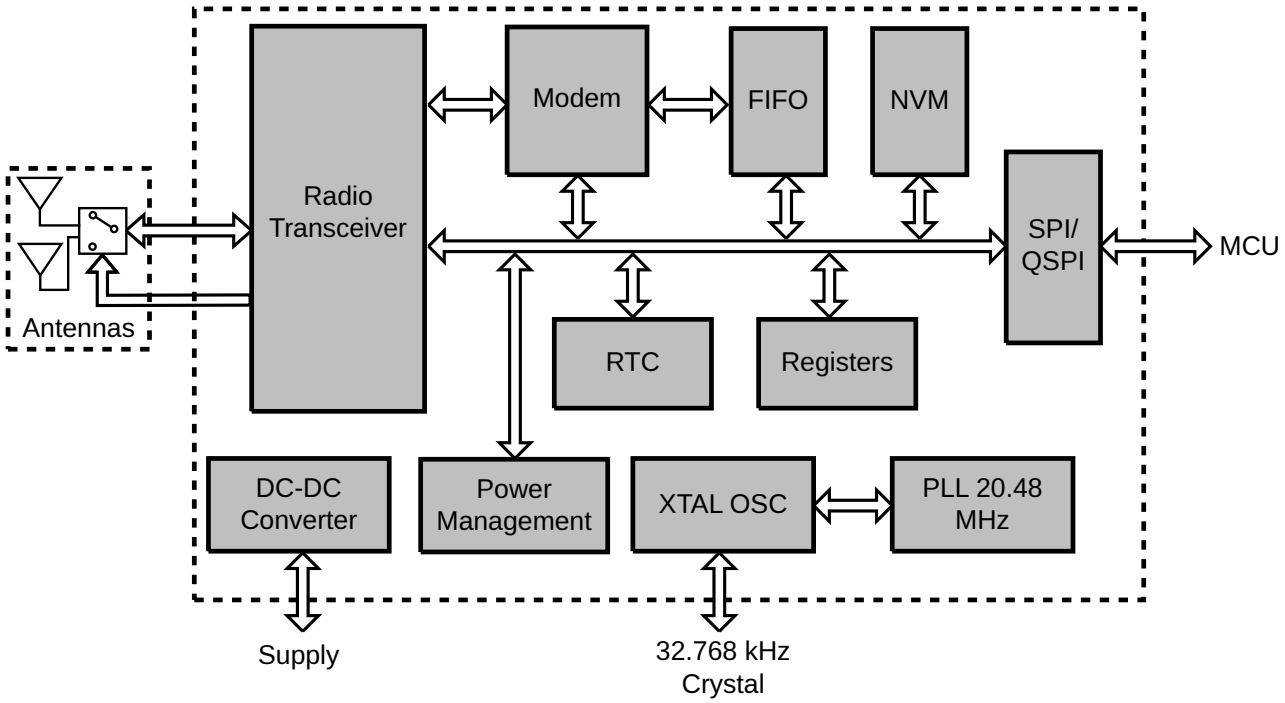


Figure 1.1: Block diagram of the SR1120.

1.2 Radio transceiver description

The radio transceiver supports a proprietary impulse radio and a standard IEEE-802.15.4ab draft compliant radio. The transceiver operates in the license-free UWB spectrum. The transmitter is highly flexible and can be adapted to efficiently fill the available RF spectrum, taking into account the realized antenna gains in a specific implementation. The receiver operates in non-coherent mode and is robust to in-band and out-of-band interferers. The transceiver supports multiple PHY rates to enhance application requirements. On-Off Keying (OOK) is supported by the modem, as well as differential encoding (2 bit-PPM) , configurable punctured FEC , variable preamble length, variable sync word and programmable CRC polynomial. A 2kbit receive and transmit buffer are available to buffer the packets. The data rate and available RF spectrum constrains the allowable transmitter parameters. As a result, the link budget is higher for low data rate links and lower for high data rate links. Additionally, the transceiver operates with only a 32.768 kHz quartz, enabling very low power timing. The transmitter can be programmed to have a variable pulse width via radio control settings. The transmitter can also be controlled to vary the number of pulses used within each symbol. A variable output power setting can be fine-tuned during each transmission.

1.3 Clocks

A 32.768 kHz clock is generated by the XTAL oscillator using an external crystal. This clock powers the Timer implementation. A 40.96 MHz clock is synthesized on-chip from the XTAL clock using a PLL. This PLL requires an off-chip loop filter and is power cycled according to the usage profile.

1.4 Power modes

The SR1120 supports several power modes. Accurate power cycling is key to reaching high energy efficiencies. The sleep controller contains automatic sleep and wake-up conditions.

Function	Shutdown	Deep Sleep	Shallow Sleep	Idle	Active
Register retention	No	Yes	Yes	Yes	Yes
SPI communication	Off	On	On	On	On
PLL clock	Off	Off	On	On	On
DC-DC output	Off	Off	Off	On	On
Radio	Off	Off	Off	Off	On
Next State	*Deep Sleep*	*Shallow Sleep*	*Idle*	*Active*	*Any*

Table 1.2: Power States

1.5 Reset

At system startup, the `RSTN` pin should be held low until the supply voltage is stable. Normal operation starts after pulling `RSTN` high.

The Shutdown mode allows minimum current consumption, and can be entered by pulling `SHUTDOWN` high. Wake-up from Shutdown mode is accomplished by pulling `SHUTDOWN` low, while keeping `RSTN` low to reset the SR1120. Normal operation starts after pulling `RSTN` high.

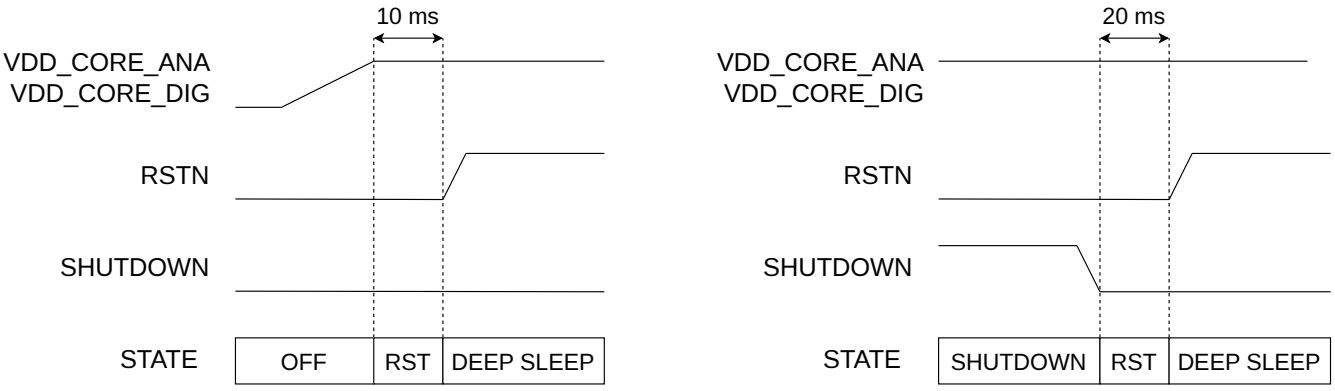


Figure 1.3: Timing sequence during power on (left) and shutdown (right).

Figure 1.3 illustrates the power-on sequence and wake-up out of shutdown. Note that in both sequences the digital core is initialized to the default register state.

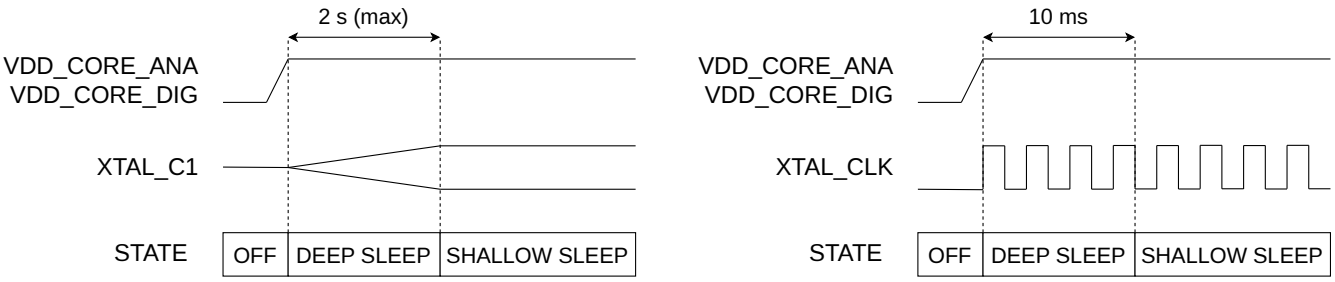


Figure 1.4: XTAL sequence with crystal on XTAL pins (left) or an external digital clock on XTAL_CLK (right).

After power-on or leaving shutdown mode, the XTAL clock source has to provide a stable 32.768 kHz clock before deep sleep can be exited into shallow sleep. An external 32.768 kHz clock source can be used as an input to the `XTAL_CLK` pin, as shown in Figure 1.4.

1.6 Sleep Timing

Depending on the packet rate, the radio can transition in either the deep sleep, shallow sleep or idle state in between packets. The figure below illustrates the state transitions in these three modes.

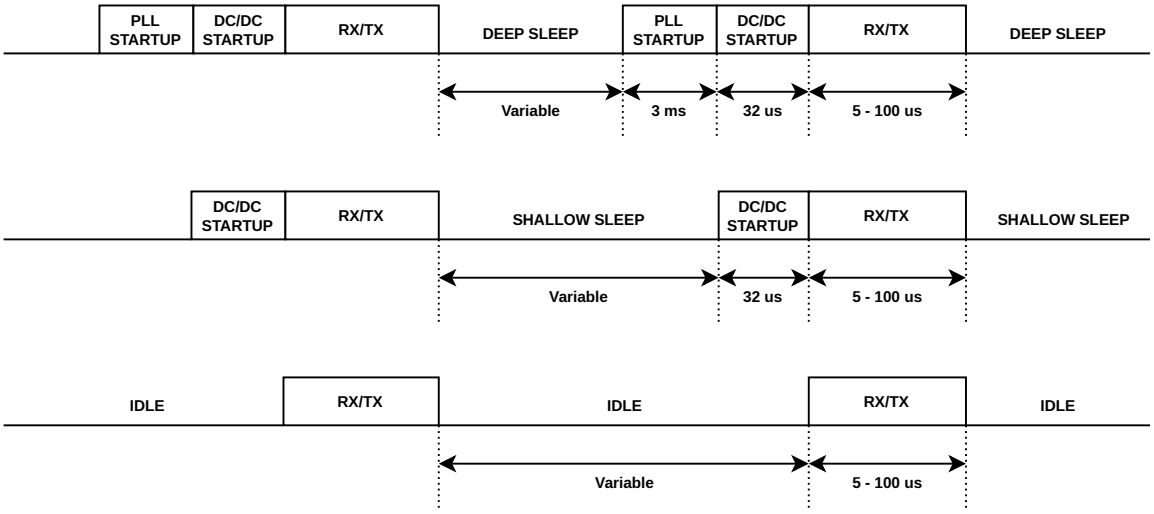


Figure 1.5: State timing profiles (at 3.3 V) for deep sleep (top), shallow sleep (middle) and idle (bottom).

1.7 SPI QSPI interface

The SR1120 can be controlled via a SPI/QSPI interface which can operate up to a frequency of 40 MHz. After reset the interface is set to be in SPI mode (SPI mode 0). The following transaction examples are supported in SPI and QSPI mode. A transaction is determined by the `CSN` signaling. A transaction starts with `CSN` going low and finishes when it goes up. The bit transfer level on SPI is MSB first, see Figure 1.6.

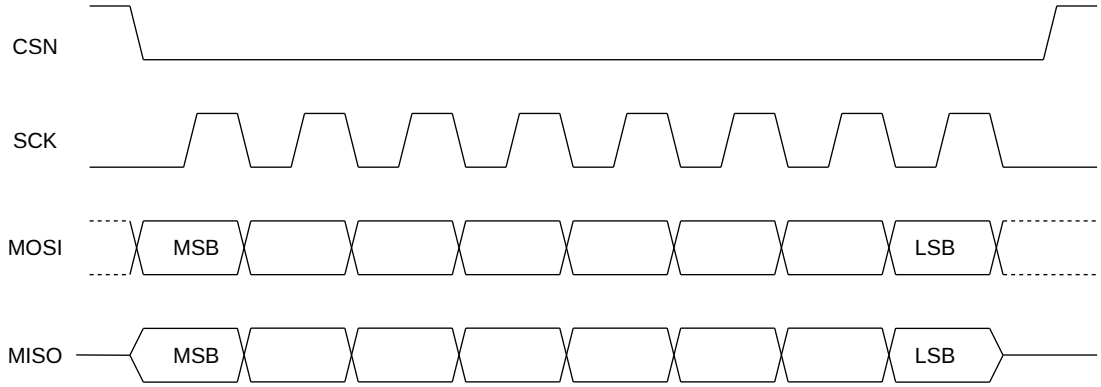


Figure 1.6: Timing of a single byte on the SPI protocol.

The non-burst (normal) read and write register protocol is illustrated in Figure 1.8. Every read and write transaction starts with a command byte, where the bit 7 indicates if a normal (0) or a burst (1) transfer is coming and the bit 6 indicates if the transaction will be a reading (0) or writing (1) operation. The 6 remaining bits constitute the register address to access for that transaction. During the data transfer of a read transaction, the contents at the designated register address are returned on `MISO` while the `MOSI` signal values is ignored.

Command bit	Description	Value
7	Mode	*1* burst, *0* normal (non-burst)
6	Operation (Read or Write)	*1* write, *0* read
5:0	Address	Value

Table 1.7: Command bit mapping

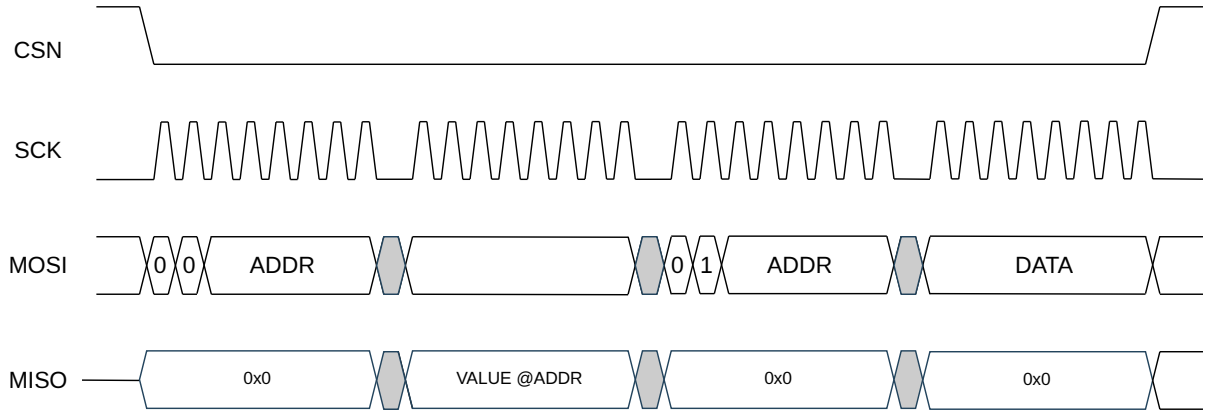


Figure 1.8: SPI read and write register transaction.

During the data transfer of a write transaction, the new value to write at the designated register address is received through the `MOSI` signal while the `MISO` signal remains low.

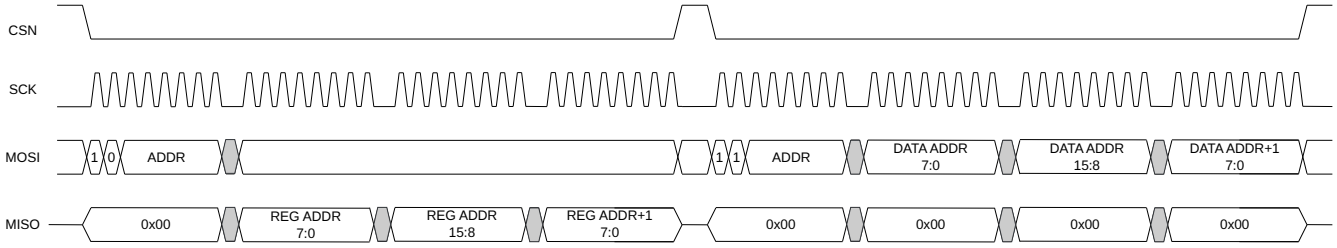


Figure 1.9: SPI burst read and write transaction.

The SPI burst transfers are illustrated in Figure 1.9. SPI burst transfers continuously transfer data bits every SPI clock cycle indefinitely until `CSN` is raised. The address will be incremented while the transaction is active for every two (2) bytes.

QSPI uses the same mechanism that is offered by SPI, however instead of using `MISO` / `MOSI` the QSPI uses all four (4) bits for data transmission (IO3,IO2,IO1,IO0), see Figure 1.10. The device changes IO direction, on the data bytes, depending if a read operation transaction is issued.

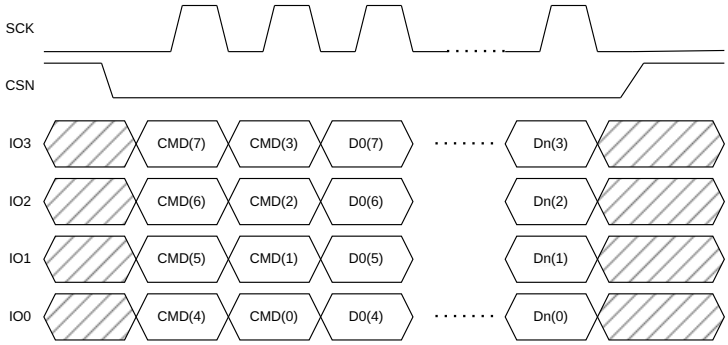


Figure 1.10: QSPI transaction.

1.8 Interrupt controller and Timer

The interrupt controller can be programmed to launch an asynchronous interrupt A on IRQ on certain events. This allows the MCU to sleep. Common interrupt events include packet reception, frame transmit completed, wakeup from sleep, and more.

A low-power Timer is available in every mode except shutdown, which can wake up into higher sleep modes.

1.9 NVM

A 1 kbit Non-Volatile Memory (NVM) is available for storing calibration data.

2 PHY Layer Overview

2.1 General

The PHY layer is based on impulse radio signaling in the 6.7 - 9.2 GHz frequency band. Each radio is free to fill this spectrum, as long as local UWB spectral regulations are met. The aim is to transmit at a fast rate, then release the channel as soon as possible to enhance multi-user co-existence. This type of bursty transmission also greatly reduces power consumption as compared to time-spread UWB standards.

The instantaneous bandwidth of the channel is 500 MHz to facilitate sufficient sensitivity at the receiver. The radio can transmit multiple packets at different RF frequencies in a 1 ms time frame, in order to fill more spectrum and increase the energy in a symbol. Information is encoded in the existence or nonexistence of a pulse train.

On-Off Keying (OOK) is supported. The payload and CRC are optionally encoded with a 1/1.25, 1/1.375, 1/1.5, 1/1.625, 1/1.75, 1/1.875 or 1/2.0 Viterbi rate, punctured convolutional Forward Error Correction (FEC) code.

2.2 Symbol Structure

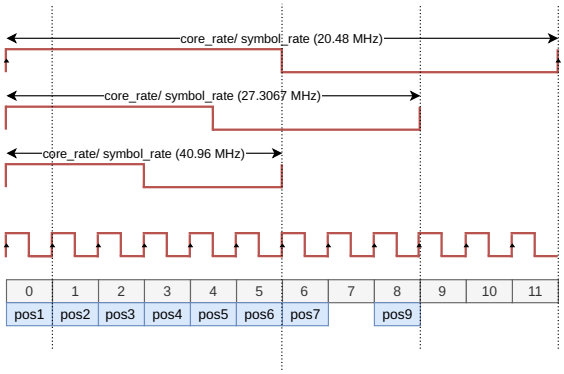


Figure 2.1: Symbol Structure

Figure 2.1 outlines the structure of a symbol. Each symbol has a length of PHY period, and is subdivided in pulse positions of 4.069 ns. Positions 1 to 6 and 7,9 can be controlled for pulse patterning.

2.3 Encoding

Base	Data	0	1		
	Symbol	0	1		
1-bit Differential	Data	0	1		
	Symbols	10	01		
2-bit Differential	Data	00	01	11	10
	Symbols	1000	0100	0010	0001

Figure 2.2: Encoding Formats

Figure 2.2 outlines the available encoding formats. In the base modulation, the total absence of pulses encodes a 0 bit, and the existence of pulses encodes a 1 bit. Both preamble and sync word are encoded by the base modulation only.

In addition, the payload and CRC can be optionally encoded in a half rate 1-bit or 2-bit differential encoding. 1-bit differential encoding represents bits in two OOK modulated symbols, with a zero bit encoded as a 1 followed by a 0, and a one bit encoded as a 0 followed by a 1. 2-bit differential encoding represents two bits in four OOK modulated symbols, one of which carries a 1 symbol and the other 3 carry a 0 symbol.

2.4 Packet Structure

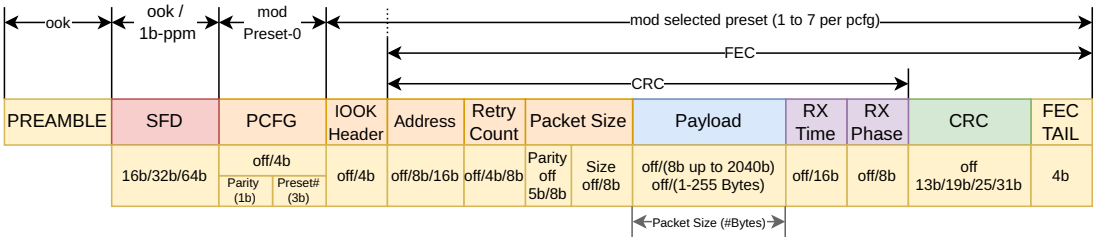


Figure 2.3: Packet Structure

Figure 2.3 outlines the structure of a packet. Each packet consists of a preamble and sync word (SFD). The optional fields are inserted into the packet structure and do not consume any bits of the Payload.

Field	Availability	Description
Preamble	Always	Predefined alternating sequence of 0 and 1 symbols
SFD (Sync. word)	Always	Start of frame delimiter
PCFG	If PCFG enabled	Preset PHY configuration
IOOK Header	If modulation = IOOK	Indicates the polarity used for IOOK
Address	If field enabled	Addressing for packet filtering
Retry Count	If field enabled	Number of times the transmission of the current frame was delayed due to CCA fail
Packet Size	If field enabled	Variable size of payload in bytes
Payload	Always unless null	Data bytes
RX Time	If field enabled; auto-reply frame only	Timer value when the previous standalone frame was received by TX
RX Phase	If field enabled; auto-reply frame only	Phase tracking value when the previous standalone frame was received by TX
CRC	If CRC enabled	CRC remainder to validate transfer integrity
FEC Tail	If FEC enabled	Necessary to complete the FEC code

Table 2.4: Field Availability and Description

If the address field is enabled, packet filtering is performed in the PHY layer based on the received address. If the packet size field is enabled, the receiver will put that exact amount of bytes into the FIFO. In default broadcast, these fields are disabled and the default packet size is 16 bytes.

3 Pinout

3.1 Package

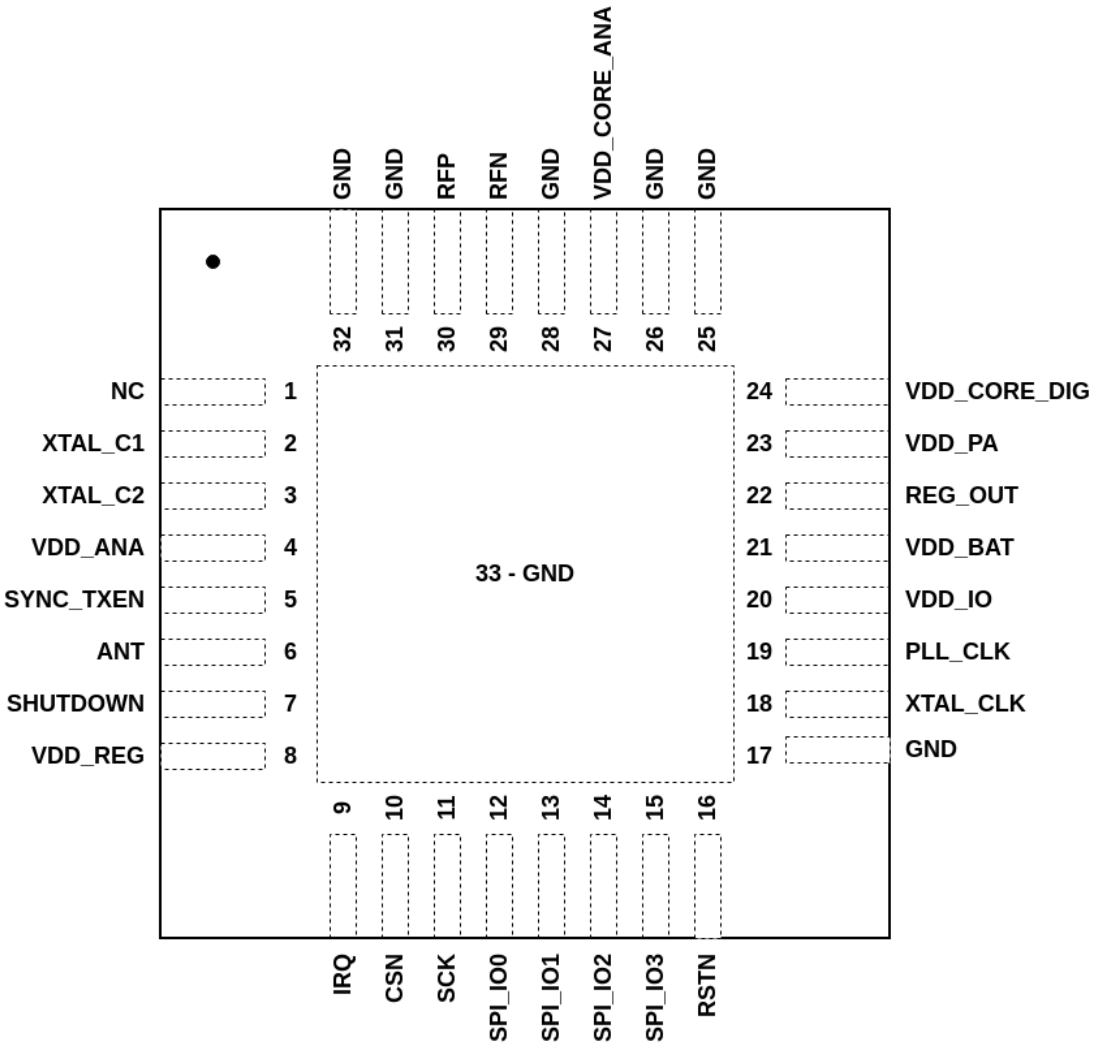


Figure 3.1: QFN32 Package (top view)

3.2 Pinouts

Name	Pin	Type	Description
RFP	30	Analog Input/Output	Positive RF input/output.
RFN	29	Analog Input/Output	Negative RF input/output.
CSN	10	Digital Input	SPI Chip Select.
SCK	11	Digital Input	SPI clock.
SPI_IO0/MOSI	12	Digital Input/Output	SPI data 0 or MOSI.
SPI_IO1/MISO	13	Digital Input/Output	SPI data 1 or MISO.
SPI_IO2	14	Digital Input/Output	SPI data 2.
SPI_IO3	15	Digital Input/Output	SPI data 3.
XTAL_C1	2	Analog Input	32.768 kHz crystal oscillator pin 1.
XTAL_C2	3	Analog Output	32.768 kHz crystal oscillator pin 2.
NC	1	No connect	
SYNC_TXEN	5	Digital Output	Multipurpose IO, Recommended test point connection.
ANT	6	Digital Output	Antenna switch driver.
SHUTDOWN	7	Digital Input	Normal operation when pin is low, chip is off when pin is high. Connect to GND if not used.
RSTN	16	Digital Input	Main reset pin; driving it low resets the SR1120.
IRQ	9	Digital Output	Generates user-configurable external interrupt.
PLL_CLK	19	Digital Input/Output	PLL external clock input/output.
XTAL_CLK	18	Digital Input/Output	XTAL external clock input/output.
VDD_ANA	4	Power	Supply for analog circuits, connect to VDD_BAT.
VDD_BAT	21	Power	Supply for switching/digital circuits.
VDD_IO	20	Power	Supply for IO cells.
VDD_CORE_ANA	27	Power	Regulated 1V supply to analog radio PHY.
VDD_CORE_DIG	24	Power	Regulated 1V supply to digital radio PHY.
VDD_REG	8	Power	Connect to decoupling capacitor.
VDD_PA	23	Power	Supply for PA, connect to either VDD_BAT or VDD_CORE.
REG_OUT	22	Analog Output	Output of DC-DC regulator.
GND	17,25,26,28,31,32,33	Power	All GND pins must be connected together in the lowest impedance possible. Return currents are directed through the QFN thermal pad.

Table 3.2: Pin Descriptions.

4 Application Overview

4.1 Application Circuit

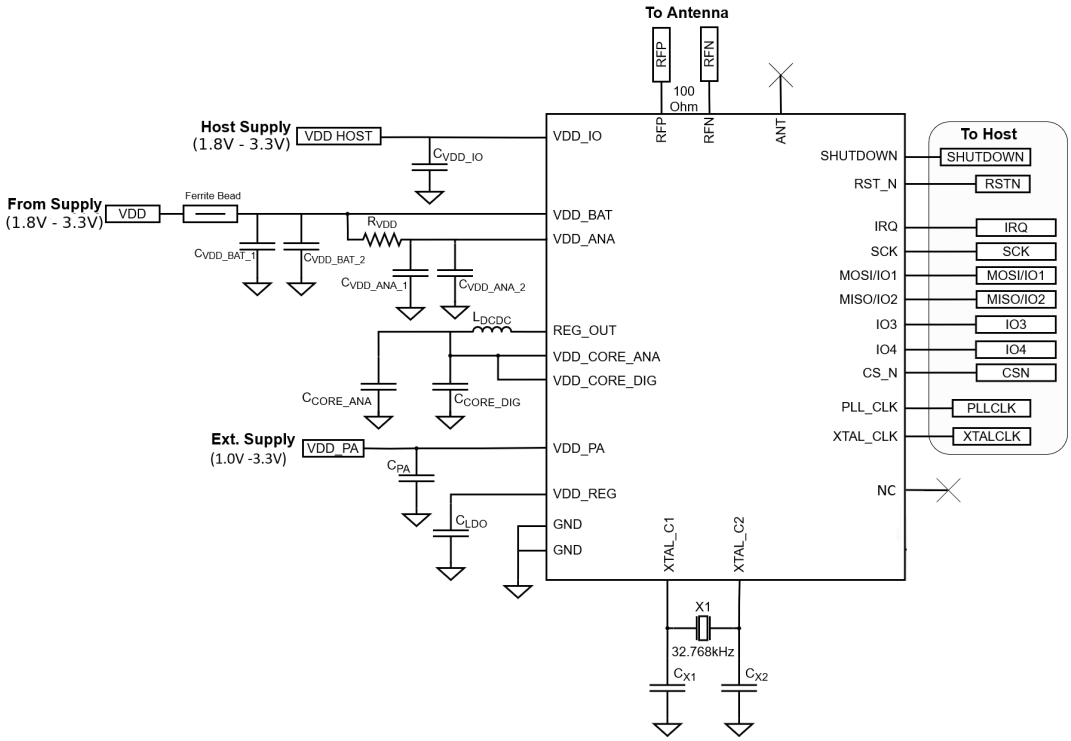


Figure 4.1: Application reference circuit

The schematic of a typical application is given in figure above.

4.2 Application Guidelines

Please observe the following guidelines:

- Connect RFP and RFN to the balun/antenna with a 100 Ohm differential transmission line.
- Connect VDD, VDD_HOST to the supply voltage with a low impedance trace.
- Place $C_{VDD_BAT_1}$ and $C_{VDD_BAT_2}$ as close as possible to VDD_BAT ($C_{VDD_BAT_1}$ and $C_{VDD_BAT_2}$ can be increased for better supply noise rejection).
- Place $C_{VDD_ANA_1}$ and $C_{VDD_ANA_2}$ as close as possible to VDD_ANA ($C_{VDD_ANA_1}$ and $C_{VDD_ANA_2}$ can be increased for better supply noise rejection).
- Connect supply to VDD_PA and place C_{PA} as close as possible to VDD_PA pin.
- Place C_{CORE_ANA} and C_{CORE_DIG} as close as possible to VDD_CORE_ANA and VDD_CORE_DIG pins.
- Place C_{LDO} as close as possible to VDD_REG pin.
- When using SPI interface, connect CS, SCK, SPI_IO0/MOSI, SPI_IO1/MISO, RSTN and IRQ to your MCU. When using QSPI interface, connect all the QSPI_IO0-3 to your MCU.
- Use of SHUTDOWN is optional. If not used, connect SHUTDOWN to GND.
- Connect all GND pins to GND. Connect all GND pads together with low impedance.

- When using XTAL_CLK and/or PLL_CLK as clock inputs, connect to your clock source. Leave unconnected otherwise.
- Minimize the capacitance on the REG_OUT node, including the parallel capacitance of the inductor.
- Adjust C_{x1} and C_{x2} to the specifications of your crystal.
- Inductor L_{DCDC} shall be < 0.75 Ω DCR, > 200 mA saturation current, and > 200 MHz self-resonance frequency.

4.2.1 Recommended Part Values

For recommended component values, please refer to the SR1120 Hardware Design Guide, which is included in the EVK Brief available for download at:

https://www.sparkmicro.com/evks/sr1120-evaluation-kit/

5 Electrical Characteristics

5.1 Maximum Ratings

PARAMETER	TEST CONDITIONS	MIN	MAX	UNIT
Storage Temperature		-50	150	℃
Pin Voltage (VDD_BAT)		-0.3	3.6	V
Pin Voltage (VDD_ANA)[1]		-0.3	3.6	V
Pin Voltage (Analog I/O)[1]		-0.3	3.6	V
Pin Voltage (VDD_IO)[2]		-0.3	3.6	V
Pin Voltage (Digital I/O)[2]		-0.3	3.6	V
Pin Voltage (VDD_PA)		-0.3	3.6	V
Pin Voltage (VDD_CORE_ANA)		-0.3	1.1	V
Pin Voltage (VDD_CORE_DIG)		-0.3	1.1	V
Pin Voltage, VDD_ANA - VDD_BAT			0.3	V
Pin Voltage(Analog I/O) - VDD_BAT			0.3	V
Pin Voltage, VDD_IO - VDD_BAT			0.3	V
Pin Voltage(Digital I/O) - VDD_BAT			0.3	V
Pin Voltage, VDD_PA - VDD_BAT			0.3	V
ESD (Human Body Model)			2500	V
ESD (Charged Device Model)			750	V

Table 5.1: Absolute Maximum Ratings.

(1)VDD_ANA supplies all analog input and output pins, as specified in Table 3.2.

(2)VDD_IO supplies all digital input and output pins, as specified in Table 3.2.

5.2 Recommended Operating Conditions

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Operating Temperature		-40		85	℃
Supply Voltage	Voltage on VDD_ANA, VDD/IO, VDD_BAT	1.8		3.3	V
Supply Voltage	Voltage on VDD_PA	1.0		3.3	V

Table 5.2: Recommended Operating Conditions.

5.3 DC characteristics

Parameter	Test Conditions	MIN	TYP	MAX	UNIT
Shutdown Current, VDD_BAT	VDD_BAT=3.3V		157		nA
	VDD_BAT=2.5V		20		nA
	VDD_BAT=1.8V		6		nA
Deep Sleep Current, VDD_BAT	VDD_BAT=3.3V		1.1		uA
	VDD_BAT=2.5V		0.77		uA
	VDD_BAT=1.8V		0.57		uA
Shallow Sleep Current, VDD_BAT	VDD_BAT=3.3V		58		uA
	VDD_BAT=2.5V		57		uA
	VDD_BAT=1.8V		57		uA
Idle Current, VDD_BAT	VDD_BAT=3.3V		390		uA
	VDD_BAT=2.5V		450		uA
	VDD_BAT=1.8V		565		uA
DLL On Current, VDD_BAT	VDD_BAT=3.3V		500		uA
	VDD_BAT=2.5V		580		uA
	VDD_BAT=1.8V		740		uA
TX Current[1,2] , VDD_BAT	VDD_BAT=3.3V		3.3		mA
	VDD_BAT=2.5V		4.2		mA
	VDD_BAT=1.8V		5.7		mA
RX Current, VDD_BAT	VDD_BAT=3.3V		9.7		mA
	VDD_BAT=2.5V		12.1		mA
	VDD_BAT=1.8V		16		mA
Shutdown Current, VDD_PA	VDD_PA=1.8V		125		nA
Deep Sleep Current, VDD_PA	VDD_PA=1.8V		0.27		uA
Shallow Sleep Current, VDD_PA	VDD_PA=1.8V		0.30		uA
Idle Current, VDD_PA	VDD_PA=1.8V		1.2		uA
DLL On Current, VDD_PA	VDD_PA=1.8V		1.2		uA
TX Current[1,2] , VDD_PA	VDD_PA=1.8V		6		mA
RX Current, VDD_PA	VDD_PA=1.8V		11		uA

Table 5.3: DC characteristics.

(1)TX current given for phy_rate = 20.48 MHz, number of pulses per symbol = 2, continuous pulse transmission (OOK all ones). Note that this is equivalent to the TX current for phy_rate = 40.96 MHz, number of pulses per symbol = 1, continuous pulse transmission (OOK all ones)

(2)TX current scales with phy_rate, number of pulses per symbol, and the density of OOK ones in the transmission.

5.4 Timing characteristics

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Reset hold time	RSTN low	10			ms
NVM startup time	After entering Idle mode from Shallow Sleep	25			us
Transition time-Shutdown to Deep sleep			20		ms
Transition time-Deep sleep to Shallow sleep			1.5		ms
Transition time-Shallow sleep to Idle			32		us
Transition time-Idle to Active			500		ns

Table 5.4: Timing characteristics.

5.5 Internal XTAL Characteristics

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Crystal oscillator frequency			32.768		kHz
Crystal frequency tolerance		-30		+30	ppm
Startup time		1.5	1.75	2	s

Table 5.5: Internal XTAL characteristics.

5.6 External XTAL Requirements

PARAMETER	TEST CONDITIONS[1]	MIN	TYP	MAX	UNIT
Clock frequency			32.768		KHz
Frequency tolerance		-30		+30	ppm
Duty cycle			50		%
Jitter[2]				2	ns (rms)

Table 5.6: External XTAL characteristics.

(1)General test conditions: VDD_BAT = 1.8 V to 3.3 V

(2)Time interval error (TIE) measurement, 10^6 clock cycles minimum. Jitter is expressed as RMS of TIE distribution.

5.7 Internal PLL Characteristics

PARAMETER	TEST CONDITIONS[1]	MIN	TYP	MAX	UNIT
PLL Output frequency			20.48		MHz
PLL Output frequency			27.3067		MHz
PLL Output frequency			40.96		MHz
PLL Startup time			1.5		ms

Table 5.7: Internal PLL characteristics.

(1)General test conditions: VDD_BAT = 1.8 V to 3.3 V

5.8 External PLL Requirements

PARAMETER	TEST CONDITIONS[1]	MIN	TYP	MAX	UNIT
Clock frequency			20.48		MHz
Clock frequency			27.3067		MHz
Clock frequency			40.96		MHz
Frequency tolerance		-30		+30	ppm
Duty cycle			50		%
Jitter[2]				2	ns (rms)

Table 5.8: External PLL characteristics.

(1)General test conditions: VDD_BAT = 1.8 V to 3.3 V

(2)Time interval error (TIE) measurement, 10^6 clock cycles minimum. Jitter is expressed as RMS of TIE distribution.

5.9 DC-DC Characteristics

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Output voltage			1.0		V
Efficiency	20 mW load		75		%
Ripple voltage	C_DCDC = 1.0 uF L_DCDC = 180 nH		10		mV

Table 5.9: DC-DC characteristics.

5.10 Radio Characteristics

5.10.1 Transmitter

Parameter	Test Conditions[1]	MIN	TYP	MAX	UNIT
TX Average Power, Symbol Energy[2,3]	num_pulse = 1, pulse_width = med, tx_power = max		-3.0		dBm / 20.48 MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-7.7		dBm / 20.48 MHz
	num_pulse = 2, pulse_width = med, tx_power = max		0.1		dBm / 20.48 MHz
	num_pulse = 2, pulse_width = med, tx_power = min		-4.6		dBm / 20.48 MHz
	num_pulse = 3, pulse_width = med, tx_power = max		2.0		dBm / 20.48 MHz
	num_pulse = 3, pulse_width = med, tx_power = min		-2.7		dBm / 20.48 MHz
TX Average Power Step Size	num_pulse = 1,2 or 3, pulse_width = med, tx_power = min to max		0.7		dB
TX Bandwidth[4]	num_pulse = 2, pulse_width = max, tx_power = max		380		MHz
	num_pulse = 2, pulse_width = med, tx_power = max		565		MHz
	num_pulse = 2, pulse_width = min, tx_power = max		770		MHz
TX Lower Sidelobe Power	num_pulse = 2, pulse_width = max, tx_power = max		-24		dBc
	num_pulse = 2, pulse_width = min, tx_power = max		-28		dBc
TX Upper Sidelobe Power	num_pulse = 2, pulse_width = max, tx_power = max		-21		dBc
	num_pulse = 2, pulse_width = min, tx_power = max		-22		dBc
Peak Power Spectral Density[3,5]	num_pulse = 1, pulse_width = med, tx_power = max		-5.9		dBm / MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-11.2		dBm / MHz
	num_pulse = 2, pulse_width = med, tx_power = max		-1.2		dBm / MHz
	num_pulse = 2, pulse_width = med, tx_power = min		-6.5		dBm / MHz
	num_pulse = 3, pulse_width = med, tx_power = max		0.4		dBm / MHz
	num_pulse = 3, pulse_width = med, tx_power = min		-4.9		dBm / MHz
Instantaneous Peak Power[3]	num_pulse = 1, pulse_width = med, tx_power = max		2.9		Vpp
			10.7		dBm
	num_pulse = 1, pulse_width = med, tx_power = min		1.9		Vpp
			6.6		dBm
TX Center Frequency		6.7		9.2	GHz

Table 5.10: Radio TX characteristics PHY_Rate = 20.48 MHz, VDD_PA = 1.8 V

(1)General test conditions: VDD_BAT = 3.3 V, VDD_PA = 1.8 V, phy rate = 20.48 MHz, frequency = 7.5 GHz, symbol pattern = OOK all ones.

(2)Symbol energy expressed as dBm / 20.48 MHz, equal to the average power integrated over an entire symbol duration (Tsymbol = 1/phy rate).

(3)Measured as differential output at chip edge (RFP pin - RFN pin) into 100 ohm balanced load.

(4)Equivalent bandwidth containing 90% of spectral energy.

(5)Peak power, resolution bandwidth = 50 MHz.

Parameter	Test Conditions[1]	MIN	TYP	MAX	UNIT
TX Average Power, Symbol Energy[2,3]	num_pulse = 1, pulse_width = med, tx_power = max		-6.5		dBm / 20.48 MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-9.8		dBm / 20.48 MHz
	num_pulse = 2, pulse_width = med, tx_power = max		-3.4		dBm / 20.48 MHz
	num_pulse = 2, pulse_width = med, tx_power = min		-6.7		dBm / 20.48 MHz
	num_pulse = 3, pulse_width = med, tx_power = max		-1.6		dBm / 20.48 MHz
	num_pulse = 3, pulse_width = med, tx_power = min		-4.9		dBm / 20.48 MHz
TX Average Power Step Size	num_pulse = 1,2 or 3, pulse_width = med, tx_power = min to max		0.5		dB
TX Bandwidth[4]	num_pulse = 2, pulse_width = max, tx_power = max		360		MHz
	num_pulse = 2, pulse_width = med, tx_power = max		545		MHz
	num_pulse = 2, pulse_width = min, tx_power = max		750		MHz
TX Lower Sidelobe Power	num_pulse = 2, pulse_width = max, tx_power = max		-18		dBc
	num_pulse = 2, pulse_width = min, tx_power = max		-25		dBc
TX Upper Sidelobe Power	num_pulse = 2, pulse_width = max, tx_power = max		-14		dBc
	num_pulse = 2, pulse_width = min, tx_power = max		-20		dBc
Peak Power Spectral Density[3,5]	num_pulse = 1, pulse_width = med, tx_power = max		-9.3		dBm / MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-13.1		dBm / MHz
	num_pulse = 2, pulse_width = med, tx_power = max		-4.6		dBm / MHz
	num_pulse = 2, pulse_width = med, tx_power = min		-8.4		dBm / MHz
	num_pulse = 3, pulse_width = med, tx_power = max		-3.3		dBm / MHz
	num_pulse = 3, pulse_width = med, tx_power = min		-7.1		dBm / MHz
Instantaneous Peak Power[3]	num_pulse = 1, pulse_width = med, tx_power = max		1.88		Vpp
			6.4		dBm
	num_pulse = 1, pulse_width = med, tx_power = min		1.48		Vpp
			4.4		dBm
TX Center Frequency		6.7		9.2	GHz

Table 5.11: Radio TX characteristics PHY_Rate = 20.48 MHz, VDD_PA = 1.0 V

(1)General test conditions: VDD_BAT = 3.3 V, VDD_PA = 1.0 V, phy rate = 20.48 MHz, frequency = 7.5 GHz, symbol pattern = OOK all ones.

(2)Symbol energy expressed as dBm / 20.48 MHz, equal to the average power integrated over an entire symbol duration (Tsymbol = 1/phy rate).

(3)Measured as differential output at chip edge (RFP pin - RFN pin) into 100 ohm balanced load.

(4)Equivalent bandwidth containing 90% of spectral energy.

(5)Peak power, resolution bandwidth = 50 MHz.

Parameter	Test Conditions[1]	MIN	TYP	MAX	UNIT
TX Average Power, Symbol Energy[2,3]	num_pulse = 1, pulse_width = med, tx_power = max		0.0		dBm / 40.96 MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-4.8		dBm / 40.96 MHz
TX Average Power Step Size	num_pulse = 1, pulse_width = med, tx_power = min to max		0.7		dB
TX Bandwidth[4]	num_pulse = 1, pulse_width = max, tx_power = max		375		MHz
	num_pulse = 1, pulse_width = med, tx_power = max		565		MHz
	num_pulse = 1, pulse_width = min, tx_power = max		770		MHz
TX Lower Sidelobe Power	num_pulse = 1, pulse_width = max, tx_power = max		-24		dBc
	num_pulse = 1, pulse_width = min, tx_power = max		-28		dBc
TX Upper Sidelobe Power	num_pulse = 1, pulse_width = max, tx_power = max		-15		dBc
	num_pulse = 1, pulse_width = min, tx_power = max		-23		dBc
Peak Power Spectral Density[3,5]	num_pulse = 1, pulse_width = med, tx_power = max		-5.2		dBm / MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-10.6		dBm / MHz
Instantaneous Peak Power[3]	num_pulse = 1, pulse_width = med, tx_power = max		2.90		Vpp
			10.7		dBm
	num_pulse = 1, pulse_width = med, tx_power = min		1.90		Vpp
			6.6		dBm
TX Center Frequency		6.7		9.2	GHz

Table 5.12: Radio TX characteristics PHY_Rate = 40.96 MHz, VDD_PA = 1.8 V

(1)General test conditions: VDD_BAT = 3.3 V, VDD_PA = 1.8 V, phy rate = 40.96 MHz, frequency = 7.5 GHz, symbol pattern = OOK all ones.

(2)Symbol energy expressed as dBm / 40.96 MHz, equal to the average power integrated over an entire symbol duration (Tsymbol = 1/phy rate).

(3)Measured as differential output at chip edge (RFP pin - RFN pin) into 100 ohm balanced load.

(4)Equivalent bandwidth containing 90% of spectral energy.

(5)Peak power, resolution bandwidth = 50 MHz.

Parameter	Test Conditions[1]	MIN	TYP	MAX	UNIT
TX Average Power, Symbol Energy[2,3]	num_pulse = 1, pulse_width = med, tx_power = max		-3.5		dBm / 40.96 MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-6.8		dBm / 40.96 MHz
TX Average Power Step Size	num_pulse = 1, pulse_width = med, tx_power =min to max		0.5		dB
TX Bandwidth[4]	num_pulse = 1, pulse_width = max, tx_power = max		375		MHz
	num_pulse = 1, pulse_width = med, tx_power = max		565		MHz
	num_pulse = 1, pulse_width = min, tx_power = max		770		MHz
TX Lower Sidelobe Power	num_pulse = 1, pulse_width = max, tx_power = max		-24		dBc
	num_pulse = 1, pulse_width = min, tx_power = max		-28		dBc
TX Upper Sidelobe Power	num_pulse = 1, pulse_width = max, tx_power = max		-15		dBc
	num_pulse = 1, pulse_width = min, tx_power = max		-23		dBc
Peak Power Spectral Density[3,5]	num_pulse = 1, pulse_width = med, tx_power = max		-8.7		dBm / MHz
	num_pulse = 1, pulse_width = med, tx_power = min		-12.5		dBm / MHz
Instantaneous Peak Power[3]	num_pulse = 1, pulse_width = med, tx_power = max		1.88		Vpp
			6.4		dBm
	num_pulse=1, pulse_width=med, tx_power=min		1.48		Vpp
			4.4		dBm
TX Center Frequency		6.7		9.2	GHz

Table 5.13: Radio TX characteristics PHY_Rate = 40.96 MHz, VDD_PA = 1.0 V

(1)General test conditions: VDD_BAT = 3.3 V, VDD_PA = 1.0V, phy rate = 40.96 MHz, frequency = 7.5 GHz, symbol pattern = OOK all ones.

(2)Symbol energy expressed as dBm / 40.96 MHz, equal to the average power integrated over an entire symbol duration (Tsymbol = 1/phy rate).

(3)Measured as differential output at chip edge (RFP pin - RFN pin) into 100 ohm balanced load.

(4)Equivalent bandwidth containing 90% of spectral energy.

(5)Peak power, resolution bandwidth = 50 MHz.

5.10.2 Receiver

Parameter	Test Conditions[1]	MIN	TYP	MAX	UNIT
RX Sensitivity[2,3]	num_pulse = 1, symbol_rate = 20.48 MHz		-81.0		dBm / 20.48 MHz
	num_pulse = 2, symbol_rate = 20.48 MHz		-80.5		dBm / 20.48 MHz
	num_pulse = 3, symbol_rate = 20.48 MHz		-79.5		dBm / 20.48 MHz
	num_pulse = 1, symbol_rate = 40.96 MHz		-79.0		dBm / 40.96 MHz
RX Center Frequency		6.7		9.2	GHz

Table 5.14: Radio Receiver characteristics.

(1)General test conditions: VDD_BAT = 3.3 V, freq = 7.5 GHz, FEC = 1.5, packet payload length = 32 bytes, modulation = iOOK.

(2)Sensitivity expressed as average power during a data symbol, whose duration Tsymbol = 1 / symbol_rate. The symbol rate is equal to the PHY rate.

(3)Sensitivity based on a received packet error rate threshold of 10%.

5.11 IO Characteristics

PARAMETER	TEST CONDITIONS[1]	MIN	TYP	MAX	UNIT
Input low-level voltage (VIL)	VDD_BAT = 3.3 V, VDD_IO = 3.3 V			0.3*VDD_IO	V
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V			0.3*VDD_IO	V
Input high-level voltage (VIH)	VDD_BAT = 3.3 V, VDD_IO = 3.3 V	0.7*VDD_IO			V
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V	0.7*VDD_IO			V
Output low-level voltage (VOL)	VDD_BAT = 3.3 V, VDD_IO = 3.3 V, IOL = 5 mA			0.1*VDD_IO	V
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V, IOL = 5 mA			0.1*VDD_IO	V
Output high-level voltage (VOH)	VDD_BAT = 3.3V, VDD_IO = 3.3V, IOH = -5 mA	0.9*VDD_IO			V
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V, IOH = -5mA	0.9*VDD_IO			V
Input low-level leakage current (IIL)	VDD_BAT = 3.3 V, VDD_IO = 3.3 V, Input pin to 0 V	-15			nA
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V, Input pin to 0 V	-3			nA
Input high-level leakage current (IIH)	VDD_BAT = 3.3 V, VDD_IO = 3.3 V, Input pin to 3.3 V			30	nA
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V, Input pin to 1.8 V			3	nA
Output low-level current (IOL)[2]	VDD_BAT = 3.3 V, VDD_IO = 3.3 V, Output pin to 3.3 V			22	mA
	VDD_BAT = 1.8V, VDD_IO = 1.8V, Output pin to 1.8V			30	mA
Output high-level current (IOH)[3]	VDD_BAT = 3.3 V, VDD_IO = 3.3 V, Output pin to 0 V	-28			mA
	VDD_BAT = 1.8 V, VDD_IO = 1.8 V, Output pin to 0 V	-32			mA

Table 5.15: IO characteristics.

(1)All measurements at 25 ℃. For VDD_IO = 3.3 V, digital drive strength (outimped) = 0. For VDD_IO = 1.8 V, digital drive strength (outimped) = 3.

(2)IOL is sink current (indicated with positive polarity)

(3)IOH is source current (indicated with negative polarity)

6 Register Description

6.1 Register 0x01

Name	Offset	Width	Access	Reset
XTAL_CLK	0	2	rw	0
CHIP_CLK	2	2	rw	0
RESERVED	4	1	rw	0
RESERVED	5	1	rw	0
CHIP_RATE	6	2	rw	0
OUTIMPED	8	2	rw	0
QSPI	10	2	rw	0
FASTMISO	12	1	rw	0
IRQPOLAR	13	1	rw	1
RESERVED	14	1	rw	0
RESERVED	15	1	rw	0

Table 6.1: Address 0x01

`XTAL_CLK`: Controls which clock source is used for the XTAL clock, and the output on the XTAL_CLK pin.

Value	Internal XTAL Osc	XTAL Clock Source	XTAL_CLK Pin State
0	Enabled	Internal	High-Z
1	Enabled	Internal	Output Internal
2	Disabled	External	Input External
3	Enabled	External	Input External

Table 6.2: XTAL_CLK

`CHIP_CLK`: Controls which clock source is used for the chip clock, and the output on the PLL_CLK pin.

Value	Internal PLL Osc	Chip Clock Source	PLL_CLK Pin State
0	Enabled	Internal	High-Z
1	Enabled	Internal	Output Internal
2	Disabled	External	Input External
3	Enabled	External	Input External

Table 6.3: CHIP_CLK

`CHIP_RATE`: Transceiver PHY chip rate.

Value	Chip Rate (MHz)
0	20.48
1	40.96
2	27.31
3	Reserved

Table 6.4: CHIP_RATE

`IRQPOLAR`: Interrupt ReQuest output pin polarity. 1: the IRQ pin is active high. 0: the IRQ pin is active low.

`FASTMISO`: Enables faster SPI response on MISO. 0: MISO changes on falling edge of SCK. 1: MISO changes as soon as data is ready.

`OUTIMPED`: Strength of digital IO driver. 2-bit value from 0 to 3, 0 is weakest driver, 3 is strongest driver.

`QSPI`: Operating mode of the SPI/QSPI interface. In mode 3, 6 dummy cycles are discarded before the first read byte following an instruction byte, and 8 dummy cycles are discarded before listening for an instruction byte following a data byte, or a falling edge on the `CSN` pin.

Value	Mode	Dummy Cycles	SPI_IO0	SPI_IO1	SPI_IO2	SPI_IO3
0	SPI	N/A	Input	Output	High Z	High Z
1	QSPI	0	I/O	I/O	I/O	I/O
2	QSPI	1	I/O	I/O	I/O	I/O
3	QSPI	6 + 8	I/O	I/O	I/O	I/O

Table 6.5: QSPI

6.2 Register 0x02

Name	Offset	Width	Access	Reset
DLTUNING	0	5	rw	0x10
DL_LAGS	5	1	ro	0
IREFPINE	5	1	wo	0
IREFTUNE	6	6	rw	0x20
VREFTUNE	12	4	rw	0x08

Table 6.6: Address 0x02

`IREFTUNE`: Adjustment of transceiver reference current. Read from chip NVM and copy in this field.

`VREFTUNE`: Adjustment of transceiver reference voltage. Read from chip NVM and copy in this field.

`DLTUNING`: 5-bit tuning of transceiver Delay Line. Adjust until `DL_LAGS` flips from 1 to 0.

`DL_LAGS`: Read-only bit to determine Delay Line locking. 0: Delay Line is early. 1: Delay Line is late.

6.3 Registers 0x04-0x05

Name	Offset	Width	Access	Reset
CCAONTIME	0	4	rw	0x01
MAXRETRY	4	4	rw	0
TXANYWAY	8	1	rw	1
IGNORPKT	9	1	rw	1
CCAINTERV	10	6	rw	0x03

Table 6.7: Address 0x04

Name	Offset	Width	Access	Reset
CCATHRES	0	7	rw	0x50
RESERVED	7	1	rw	0
RESERVED	8	7	rw	0

Table 6.8: Address 0x05

`MAXRETRY`: Maximum number of Clear Channel Assessment retries, 4-bit value.

`CCAONTIME`: Duration of CCA check, 4-bit value, expressed in number of chip cycles = (`CCAONTIME` + 1) X 8.

`CCAINTERV`: Interval in between CCA checks, 6-bit value, expressed in number of chip cycles = (`CCAINTERV` + 1) X 32.

`IGNORPKT`: Block packet reception during CCA checks. 1: Do not trigger on packet during CCA, 0: Attempt to trigger on packet during CCA. Reserved: default to 1.

`TXANYWAY`: Action to be taken after `MAXRETRY` number of CCA checks have failed. 1: Transmit packet anyway. 0: Do not transmit.

`CCATHRES`: Decision threshold for CCA checks. Values range from 0 to 115, with lower values indicating stronger signals, and higher values indicating weaker signals. Each step represents about 0.5 dB in signal power. Recommended value: 70.

6.4 Register 0x07

Name	Offset	Width	Access	Reset
INTEGGAIN	0	4	rw	0x08
RESERVED	4	3	rw	0x04
RESERVED	7	1	rw	0
RESERVED	8	4	rw	0x0C
LNA_FREQ	12	4	rw	0x08

Table 6.9: Address 0x07

INTEGGAIN: Adjust optimal integrator gain before ADC, 4-bit value. Adjust based on PHY chip rate and number of pulses transmitted per symbol by the transmitter.

Chip Rate (MHz)	TX Pulse Count	INTEGGAIN Value
20.48	1	10
20.48	2	8
20.48	3	7
40.96	1	13
40.96	2	10
40.96	3	9
27.31	1	11
27.31	2	9
27.31	3	8

Table 6.10: INTEGGAIN

LNA_FREQ: Frequency tuning of the receiver LNA, 4-bit value. Adjust based on channel center frequency, codes 0xB to 0xF are reserved.

Value	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA
Freq (GHz)	9.09	8.60	8.36	7.95	7.70	7.45	7.21	7.05	6.68	6.43	6.31
Band	222	210	204	194	188	182	176	172	163	157	154

Table 6.11: LNA Frequency Tuning

6.5 Registers 0x08-0x09

Name	Offset	Width	Access	Reset
RXBANDFRE	0	6	rw	0x20
CFG1FREQ	8	6	rw	0x20

Table 6.12: Address 0x08

Name	Offset	Width	Access	Reset
CFG2FREQ	0	6	rw	0x20
CFG3FREQ	8	6	rw	0x20

Table 6.13: Address 0x09

RXBANDFRE: VCRO code used during reception, tuning the center frequency band, 6-bit value.

CFG1FREQ, CFG2FREQ, CFG3FREQ: VCRO code used during transmission, tuning the center frequency band, 6-bit value. Separate fields for three pulse configuration.

6.6 Registers 0x0A

Name	Offset	Width	Access	Reset
CFG1WIDTH	0	3	rw	0x04
CFG2WIDTH	4	3	rw	0x04
CFG3WIDTH	8	3	rw	0x04
TX_POWER	12	3	rw	0x04
RANDPULS	15	1	rw	1

Table 6.14: Address 0x0A

`CFG1WIDTH`, `CFG2WIDTH`, `CFG3WIDTH`: Control of transmitted pulse width, 3-bit value. Lower value is shorter pulse (wider TX bandwidth), higher value is longer pulse (narrower TX bandwidth). Separate fields for three pulse configuration.

`TX_POWER`: Amplitude control of transmitted pulse, 3-bit value. Lower value is higher amplitude, higher value is lower amplitude.

`RANDPULS`: Randomizes the phase of each pulse. Reserved: default to 1.

6.7 Registers 0x0B

Name	Offset	Width	Access	Reset
POS1PULSE	0	2	rw	0
POS2PULSE	2	2	rw	0x01
POS3PULSE	4	2	rw	0
POS4PULSE	6	2	rw	0x01
POS5PULSE	8	2	rw	0
POS6PULSE	10	2	rw	0
POS7PULSE	12	2	rw	0
POS9PULSE	14	2	rw	0

Table 6.15: Address 0x0B

`POS1PULSE`, `POS2PULSE`, `POS3PULSE`, `POS4PULSE`, `POS5PULSE`, `POS6PULSE`, `POS7PULSE`, `POS9PULSE`: For pulse positions 1-9 (excluding 8) in each transmitted symbol, `POSxPULSE` is a 2-bit value determining the content of that pulse position:

Value	Pulse Position Content
0	No Pulse
1	Pulse Configuration 1
2	Pulse Configuration 2
3	Pulse Configuration 3

Table 6.16: `POSxPULSE`

6.8 Registers 0x0C-0xD

Name	Offset	Width	Access	Reset
SLPPERIOD[15-0]	0	16	rw	0x07FF

Table 6.17: Address 0x0C

Name	Offset	Width	Access	Reset
SLPPERIOD[23-16]	0	8	rw	0
PWRUPDLAY	8	8	rw	0

Table 6.18: Address 0x0D

`SLPPERIOD`: Length of sleep period, 24 bits, in number of clock cycles of the sleep clock (chip clock in Idle Sleep, XTAL clock in Deep and Shallow sleep.)

`PWRUPDLAY`: Delay after wake-up, before starting the transmitter or receiver, 8 bits. The duration of delay in number of chip cycles is `PWRUPDLAY` X 8.

6.9 Register 0x0E

Name	Offset	Width	Access	Reset
RESERVED	0	2	rw	0
RESERVED	2	1	rw	0
TIMEOUT	3	13	rw	0

Table 6.19: Address 0x0E

`TIMEOUT`: Duration of receiver listening period before timing out, 13 bits. The duration length in number of chip cycles is `TIMEOUT` X 8 + 1.

6.10 Register 0x0F

Name	Offset	Width	Access	Reset
SYNTIMEO	0	1	rw	0
SYNRXEND	1	3	rw	0
SYNRXSTA	4	1	rw	0
SYNTXEND	5	1	rw	0
SYNWAKUP	6	1	rw	0
AUTOWAKE	7	1	rw	0
SLPTIMEO	8	1	rw	0
SLPRXEND	9	3	rw	0
SLPCCAFA	12	1	rw	0
SLPTXEND	13	1	rw	0
SLPDEPTH_WAKEONCE	14	2	rw	0x03

Table 6.20: Address 0x0F

`AUTOWAKE`: Automatically wake-up when sleep period is reached on the active sleep timer. 1: Enabled. 0: Disabled.

`SYNWAKUP`: Reset both chip and XTAL sleep timer on wake-up. 1: Enabled. 0: Disabled.

`SYNTXEND`: Reset both chip and XTAL sleep timer on end of packet transmission. 1: Enabled. 0: Disabled.

`SYNRXEND`: Reset both chip and XTAL sleep timer on end of packet reception, 3 bits. 001: Sync on correct CRC match. 000: Disabled all other reserved.

`SLPDEPTH_WAKEONCE`: Determines sleep level used when going to sleep. If `SLPDEPTH_WAKEONCE` > 0, Wake-Once is activated and only one wake-up event is triggered until `SLP_PERIOD` is overwritten again.

Value	Sleep Level	Sleep Timer	Wake-Once
0	Idle sleep	Chip clock	Off
1	Idle sleep	Chip clock	On
2	Shallow sleep	XTAL clock	On
3	Deep sleep	XTAL clock	On

Table 6.21: SLPDEPTH_WAKEONCE

`SLPTXEND`: Go to sleep after packet transmission. 1: Enabled. 0: Disabled.

`SLPRXEND`: Go to sleep after packet reception, 3 bits. 001: Sync on correct CRC match. 000: Disabled all other reserved.

`SLPTIMEO`: Go to sleep after reception timeout. 1: Enabled. 0: Disabled.

6.11 Register 0x10

Name	Offset	Width	Access	Reset
TIMEOUTE	0	1	wo	1
CRCPASSE	1	1	wo	0
BRDCASTE	2	1	wo	0
ADDRMATE	3	1	wo	0
RXENDE	4	1	wo	1
TXENDE	5	1	wo	1
ARRXENDE	6	1	wo	1
ARTXENDE	7	1	wo	1
WAKEUPE	8	1	wo	0
XOTIMERE	9	1	wo	0
CCAFAILE	10	1	wo	1
BUFTHRES	11	5	wo	0
TIMEOUTI	0	1	ro	1
CRCPASSI	1	1	ro	0
BRDCASTI	2	1	ro	0
ADDRMATI	3	1	ro	0
RXENDI	4	1	ro	1
TXENDI	5	1	ro	1
ARRXENDI	6	1	ro	1
ARTXENDI	7	1	ro	1
WAKEUPI	8	1	ro	0
XOTIMERI	9	1	ro	0
CCAFAILI	10	1	ro	1
BUFLOADI	11	1	ro	1
RXUDRFLI	12	1	ro	1
RXOVRFLI	13	1	ro	1
TXUDRFLI	14	1	ro	1
TXOVRFLI	15	1	ro	1

Table 6.22: Address 0x10

Status flags for interrupts, and enable control for interrupts asserting the IRQ pin.

Flag	IRQ Pin Enable	Raised by Event	Cleared by
TXENDI	TXENDE	End of packet transmission	Register read
RXENDI	RXENDE	End of packet reception	Register read
ARTXENDI	ARTXENDE	End of Auto-Reply transmission	Register read
ARRXENDI	ARRXENDE	End of Auto-Reply reception	Register read
CRCPASS	CRCPASS	Successful CRC pass	Register read
TIMEOUTI	TIMEOUTE	Timeout during reception	Register read
WAKEUPI	WAKEUPE	Wake up from sleep	Register read
XOTIMERI	XOTIMERE	Sleep period on XTAL timer reached	Register read
CCAFAILI	CCAFAILE	CCA check fail	Register read
ADDRMATI	ADDRMATE	Address field match	Register read
BRDCASTI	BRDCASTE	Broadcast field match	Register read
TXOVRFLI	not DISABUFI	TX FIFO overflow	Reset TX FIFO
RXOVRFLI	not DISABUFI	RX FIFO overflow	Reset RX FIFO
TXUDRFLI	not DISABUFI	TX FIFO underflow	Reset TX FIFO
RXUDRFLI	not DISABUFI	RX FIFO underflow	Reset RX FIFO
BUFLOADI	BUFTHRES > 0	RX FIFO filled above threshold	Pull from RX FIFO until below threshold

Table 6.23: IRQ Control

BUFTHRES: RX FIFO threshold, 5 bits. If the number of bytes in the RX FIFO exceeds BUFTHRES X 8, the BUFLOADI interrupt will be raised.

6.12 Register 0x11

Name	Offset	Width	Access	Reset
RESERVED	0	3	wo	0
RESERVED	3	1	rw	0
RESERVED	4	1	rw	0
RSSIONLY	5	1	rw	0
RPLYTXEN	6	1	rw	0
RADIODIR	7	1	rw	1
RESERVED	8	8	ro	0

Table 6.24: Address 0x11

RADIODIR: Selects transmission or reception mode. 1: Reception. 0: Transmission.

RPLYTXEN: Enables Auto-Reply on packets.

RSSIONLY: Enables energy detection mode on reception, tracking energy with RSSI, without packet reception.

6.13 Register 0x12

Name	Offset	Width	Access	Reset
RSSI	0	7	ro	0
RNSI	8	7	ro	0

Table 6.25: Address 0x12

`RSSI`: Signal strength indicator, latched after syncword detection, 7-bits. Values range from 0 to 115, with lower values indicating stronger signals, and higher values indicating weaker signals. Each step represents about 0.5 dB in signal power.

`RNSI`: Noise strength indicator, 7-bits. Values range from 0 to 115, with lower values indicating stronger noise, and higher values indicating weaker noise. Each step represents about 0.5 dB in noise power.

6.14 Register 0x13-0x14-0x15

Name	Offset	Width	Access	Reset
RXSFDTIME	0	16	ro	0

Table 6.26: Address 0x13

Name	Offset	Width	Access	Reset
TXSFDTIME	0	16	ro	0

Table 6.27: Address 0x14

Name	Offset	Width	Access	Reset
IRQTIME	0	16	ro	0
DISABUFI	0	1	wo	0

Table 6.28: Address 0x15

`RXSFDTIME`: Number of chip cycles between receiver wake-up and syncword detection, 16 bits.

`TXSFDTIME`: Number of chip cycles between transmitter wake-up and syncword transmission, 16 bits.

`IRQTIME`: Number of chip cycles between receiver or transmitter wake-up and assertion of the IRQ pin, 16 bits.

`DISABUFI`: Disables the FIFO underflow and overflow interrupts. 1: Disabled. 0: Enabled.

6.15 Register 0x18

Name	Offset	Width	Access	Reset
FEC_RATE0	0	3	rw	0x03
CHIPCODE0	3	3	rw	0x02
CHIPREPE0	6	2	rw	0
ISIMITIG0	8	2	rw	0
RPLYADD0	10	1	rw	0
EXPECRP0	11	1	rw	0
RESERVED	12	3	rw	0x03
RESERVED	15	1	rw	0

Table 6.29: Address 0x18

`CHIPCODE0`: Modulation used during reception and transmission, 3 bits.

Value	Modulation
0	OOK
1	Reserved
2	1BIT-PPM
3	2BIT-PPM
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Table 6.30: CHIPCODE0

`FEC_RATE0`: Forward Error Coding rate, 3 bits.

Value	FEC Redundancy
0	Disable
1	1.25
2	1.375
3	1.5
4	1.625
5	1.75
6	1.875
7	2.0

Table 6.31: FEC_RATE0

`ISIMITIG0`: Number of ISI mitigation symbols.

Value	Preamble Pauses	Payload Pauses
0	0	0
1	1	0
2	2	1
3	3	2

Table 6.32: ISIMITIG0

RPLYADD0: Copy address of received packet into address field of auto-reply. 1: Enabled. 0: Disabled.

EXPECRP0: Wait for auto-reply after packet transmission. 1: Enabled. 0: Disabled.

6.16 Register 0x2F

Name	Offset	Width	Access	Reset
PREAMBLEN	0	6	rw	0x04
SFDLENGTH	6	2	rw	0x02

Table 6.33: Address 0x2F

SFDLENGTH: Size of Start of Frame Delimiter field. Recommended value: 0.

Value	SFD Format
0	32-bit OOK
1	16-bit 1BIT-PPM
2	32-bit 1BIT-PPM
3	Reserved

Table 6.34: SFDLENGTH

PREAMBLEN: Size of preamble synchronization field. The length of the synchronization field in chip clock cycles = PREAMBLEN X 8 + 48. Recommended value: 16.

6.17 Registers 0x30-0x31

Name	Offset	Width	Access	Reset
SFDPATTRN[15-0]	0	16	rw	0xC11D

Table 6.35: Address 0x30

Name	Offset	Width	Access	Reset
SFDPATTRN[31-16]	0	16	rw	0x5EA6

Table 6.36: Address 0x31

`SFDPATTRN`: Pattern used in Start of Frame Delimiter field.

6.18 Registers 0x32-0x33

Name	Offset	Width	Access	Reset
RESERVED	0	1	rc	1
CRC_POLY[15-1]	1	15	rw	0x313A

Table 6.37: Address 0x32

Name	Offset	Width	Access	Reset
CRC_POLY[30-16]	0	15	rw	0x0372

Table 6.38: Address 0x33

`CRC_POLY`: Cyclic Redundancy Code polynomial. Recommended polynomials are given in the table below.

Polynomial	CRC field length (bits)
0x0372313b	25 bits
0x8fcc4ac9	31 bits

Table 6.39: CRC_POLY

6.19 Register 0x34

Name	Offset	Width	Access	Reset
SAVECRC	0	1	rw	0
SAVESIZE	1	1	rw	0
SAVEADDR	2	1	rw	0
SAVEPHS	3	1	rw	0
SAVETIME	4	1	rw	0
SAVERSSI	5	1	rw	0
PHSFIELD	6	1	rw	0
TIMFIELD	7	1	rw	0
OPTRANGI	8	1	rw	0
SIZESRC	9	1	rw	0
SIZEPARI	10	1	rw	0
SIZEHDR	11	1	rw	0
RETRYHDR	12	1	rw	0
ADDRLEN	13	1	rw	0
ADDRFIELD	14	2	rw	0

Table 6.40: Address 0x34

Several packet fields are additionally enabled:

`ADDRFIELD`: Enables an address field that can be used for address filtering. Control whether address matching gates packet reception, and/or auto reply transmission.

Value	Packet Address Field	Filter Packet Reception	Filter Auto-Reply Transmission
0	Disabled	No	No
1	Enabled	No	No
2	Enabled	No	Yes
3	Enabled	Yes	Yes

Table 6.41: ADDRFIELD

`ADDRLEN`: Size of the address field. 1: 16 bits. 0: 8 bits.

`SIZEHDR`: Adds an 8-bit payload size field to the packet during transmission, which is used during reception to determine the payload length. 1: Enabled. 0: Disabled.

`SIZEPARI`: Adds a 5-bit parity field to the payload size field. 1: Enable. 0: Disable.

`SIZESRC`: Selects what is used as the payload size for packet transmission. 1: Use `TXPKTSIZE`. 0: Use number of bytes in TX FIFO.

`OPTRANGI`: Reserved, set to 0.

Several of the packet fields can be automatically pushed into the RX FIFO during packet reception, in the following order:

SAVETIME: The RXSFDTIME is pushed to the RX FIFO. 1: Enabled. 0: Disabled.

SAVECHRE: The channel impulse response is pushed to the RX FIFO. 1: Enabled. 0: Disabled.

SAVERSSI: The RSSI is pushed to the RX FIFO. 1: Enabled. 0: Disabled.

SAVEADDR: The address field is pushed to the RX FIFO. 1: Enabled. 0: Disabled.

RETRYHDR: The CCA retry counter field is added as a packet field during transmission, and pushed to the RX FIFO during reception. 1: Enabled. 0: Disabled.

SAVESIZE: The payload size field is pushed to the RX FIFO. 1: Enabled. 0: Disabled.

If there is a payload field in the packet, it is saved at this position in the RX FIFO. Afterwards, the following packet fields can be pushed to the RX FIFO:

TIMFIELD: Include RXSFDTIME field in the packet during transmission. 1: Enabled. 0: Disabled.

PHSFIELD: Include phase offset field in the packet during transmission. 1: Enabled. 0: Disabled.

SAVECRC: Save the CRC remainder mismatch pattern to the RX FIFO. 1: Enabled. 0: Disabled.

6.20 Registers 0x35-0x36

Name	Offset	Width	Access	Reset
RXADDRESS	0	16	rw	0xADDB

Table 6.42: Address 0x35

Name	Offset	Width	Access	Reset
TXADDRESS	0	16	rw	0xADDA

Table 6.43: Address 0x36

RXADDRESS: Address used to do address field matching during packet reception.

TXADDRESS: Address used in address field during packet transmission.

6.21 Registers 0x37

Name	Offset	Width	Access	Reset
RXPKTSIZE	0	8	rw	0x10
TXPKTSIZE	8	8	rw	0x10

Table 6.44: Address 0x37

RXPKTSIZE: Size in bytes of payload field read during reception. If SIZESRC is 0, this register determines the maximum size read.

TXPKTSIZE: Size in bytes of payload field during transmission.

6.22 Registers 0x38

Name	Offset	Width	Access	Reset
TXBUFFER	0	8	wo	0
RXBUFFER	0	8	ro	0

Table 6.45: Address 0x38

TXBUFFER: Writing to this register pushes a byte into the TX FIFO.

RXBUFFER: Reading from this register pops a byte from the RX FIFO.

6.23 Registers 0x39-0x3A

Name	Offset	Width	Access	Reset
RXBUFLOAD	0	8	ro	0

Table 6.46: Address 0x39

Name	Offset	Width	Access	Reset
TXBUFLOAD	0	8	ro	0

Table 6.47: Address 0x3A

RXBUFLOAD: Number of bytes in the RX FIFO.

TXBUFLOAD: Number of bytes in the TX FIFO.

6.24 Registers 0x3B

Name	Offset	Width	Access	Reset
SLEEP	0	1	wo	1
INITIMER	1	1	wo	1
FLUSHRX	2	1	wo	1
FLUSHTX	3	1	wo	1
STARTTX	4	1	wo	0
RESERVED	5	1	wo	0
RESERVED	6	1	wo	0
AUTOTX	7	1	rw	0
TXRETRIES	0	4	ro	0

Table 6.48: Address 0x3B

Writing to this register starts the following actions:

`AUTOTX`: Start transmission every time after waking up from sleep.

`STARTTX`: Start transmission once after waking up from sleep.

`FLUSHTX`: Empty and reset the TX FIFO.

`FLUSHRX`: Empty and reset the RX FIFO.

`INITIMER`: Reset both XTAL and chip sleep timers.

`SLEEP`: Controls sleep.

Value	Description
0	Wake-up directly from sleep
1	Go to sleep when modem is not busy transmitting or receiving.

Table 6.49: SLEEP

`TXRETRIES`: Number of CCA retries received from CCA retry counter field in packet.

6.25 Registers 0x3C

Name	Offset	Width	Access	Reset
REF_EN	0	1	ro	0
PLL_EN	1	1	ro	0
DCDC_EN	2	1	ro	0
PROC_ON	3	1	ro	0
AWAKE	4	1	ro	0
RX_EN	5	1	ro	0
TX_EN	6	1	ro	0
INFRAME	7	1	ro	0

Table 6.50: Address 0x3C

`INFRAME`: Modem state of reception or transmission of packet. 1: Packet transmission or reception is ongoing, after the SFD. 0: Otherwise.

`TX_EN`: Transmitter power status. 1: Enabled. 0: Disabled.

`RX_EN`: Receiver power status. 1: Enabled. 0: Disabled.

`AWAKE`: Modem is awake. 1: Yes. 0: No.

`PROC_ON`: Modem power status. 1: Enabled. 0: Disabled.

`DCDC_EN`: DC-DC Converter power status. 1: Enabled. 0: Disabled.

`PLL_EN`: PLL power status. 1: Enabled. 0: Disabled.

`REF_EN`: Bias generator power status. 1: Enabled. 0: Disabled.

6.26 Registers 0x3E

Name	Offset	Width	Access	Reset
COUNTCODE	0	6	wo	0
CODEFREQ	0	8	ro	0

Table 6.51: Address 0x3E

`COUNTCODE`: VCRO code to be calibrated, 6-bit value. Write once to this register to start calibration.

`CODEFREQ`: VCRO frequency after calibration, 8-bit value. Frequency is (`CODEFREQ` + 256) X 20.48 MHz.

6.27 Registers 0x3F

Name	Offset	Width	Access	Reset
ROM_ADDR	0	7	wo	0
ROMPWRSW	7	1	wo	0
ROM_BYTE	0	8	ro	0

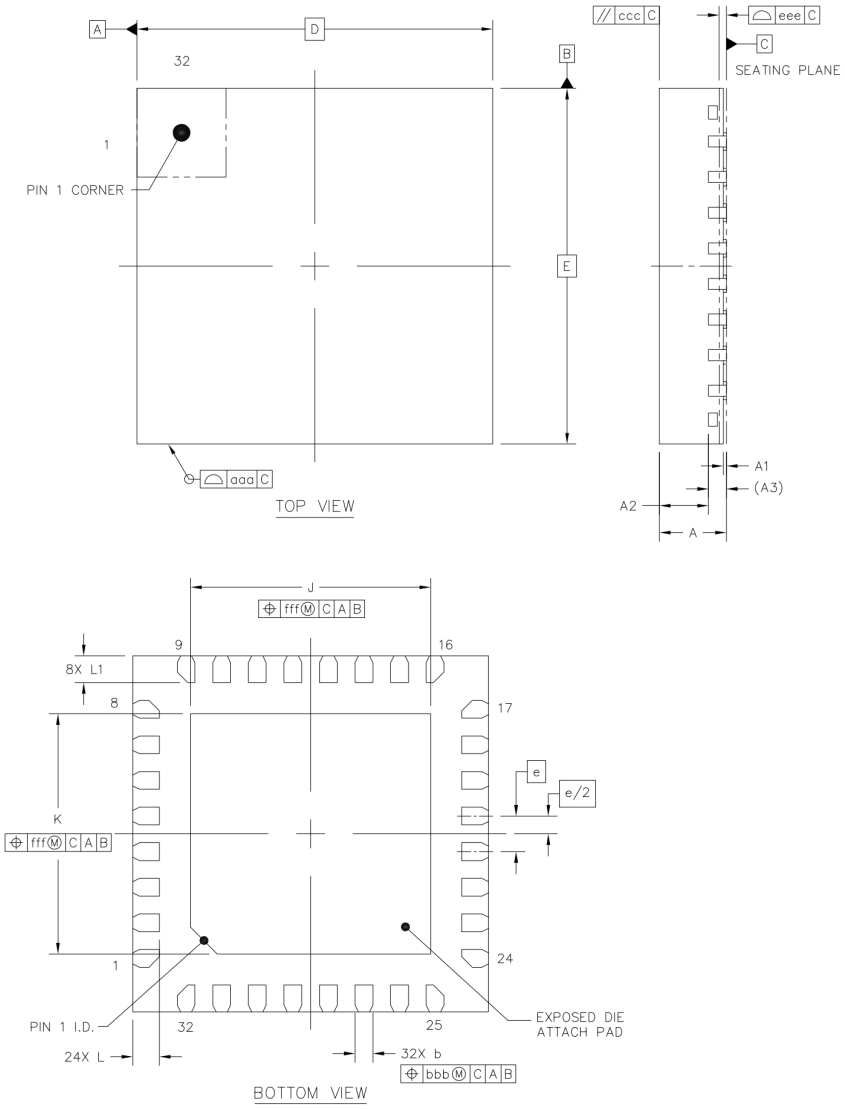
Table 6.52: Address 0x3F

ROMPWRSW: Power switch for Non-Volatile Memory (NVM). 1: Enabled. 0: Disabled.

ROM_ADDR: Address of NVM byte to be read, 7-bit value.

ROM_BYTE: Content of NVM at address location, 8-bit value.

7 Package Outline and Tape/Reel Information



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.7	0.75	0.8
STAND OFF		A1	0	0.035	0.05
MOLD THICKNESS		A2	---	0.55	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		4 BSC	
	Y	E		4 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	J	2.6	2.7	2.8
	Y	K	2.6	2.7	2.8
LEAD LENGTH		L	0.25	0.3	0.35
		L1	0.2	0.3	0.35
PACKAGE EDGE TOLERANCE		aaa		0.1	
LEAD OFFSET		bbb		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
EXPOSED PAD OFFSET		fff		0.1	

NOTES

1.0 COPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PAD.
2.0 TOTAL THICKNESS NOT INCLUDE SAW BURR.

Figure 7.1: Package Outline (Dimensions are in mm)

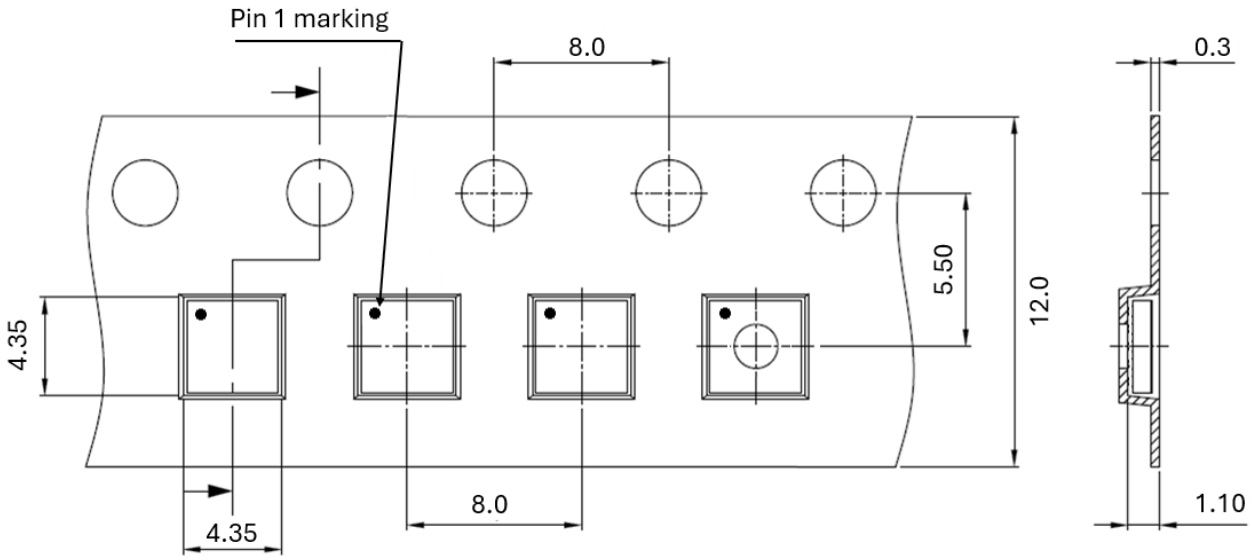


Figure 7.2: Tape and Reel (Dimensions are in mm)

8 Recommended Footprint

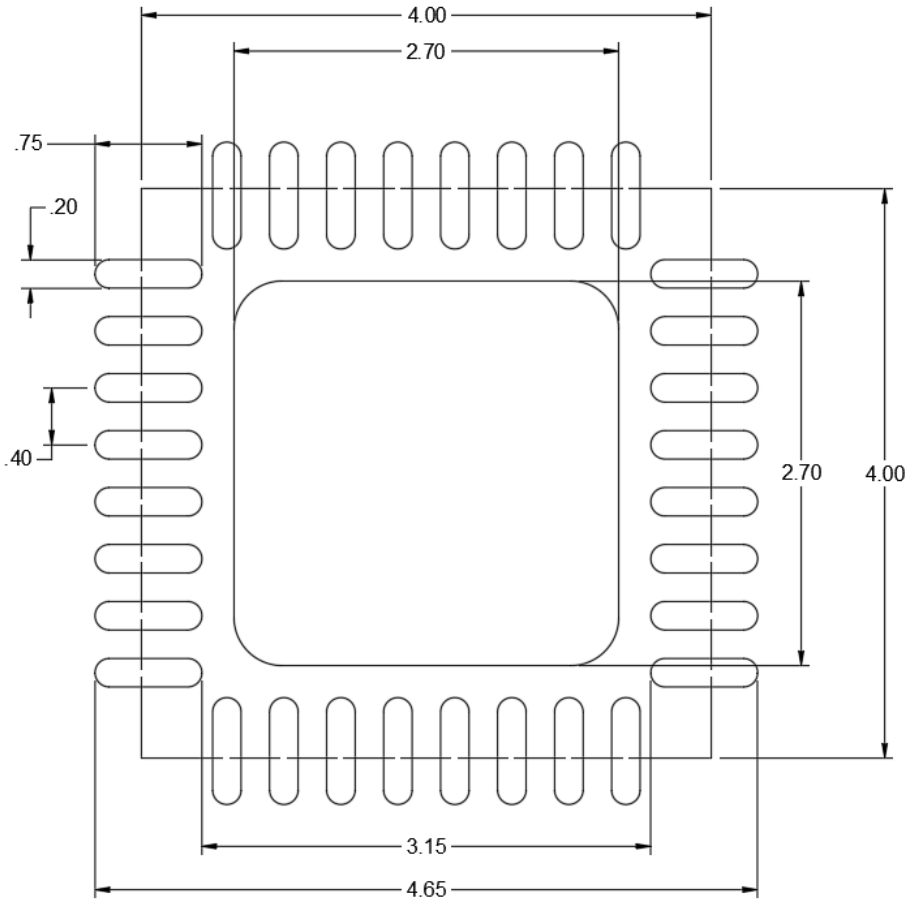


Figure 8.1: Footprint

9 Part Order Information

Part Number	Package	Size (mm)	Quantity	Carrier Format
SR1120AA-4Q32-CT	QFN32	4X4	1	Cut Tape
SR1120AA-4Q32-TY	QFN32	4X4	490	JEDEC Tray
SR1120AA-4Q32-TR	QFN32	4X4	5000	330mm Tape and Reel

Table 9.1: Part order information.

10 Revision

Revision	Date	Section	Description
2026-08-14 (1.5)	2026/08	1,2,3,4,5,7,9	Updated text and fixed typos. Included PHY layer overview. Added more radio characteristics and cleaned up application guidelines. Updated tape/reel and order information.
1.4	2025/03	1	Minor text change
1.3.3	2025/02	1,6	added data in table dc characteristics. Precise Fc.
1.3	2024/10	6	Added register file description.
1.2	2024/10	4	Updated application reference circuit, fix text.
1.1	2024/09	2,3,4	Minor text fixes, pinout, Added table in 3.2. Added table 4.1, Fix tablenote 4.2. Change min/max representation for tx_power in tables
1.0	2024/03	-	Initial release

Table 10.1: Revisions

11 Notice

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